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Follicular phase of menstrual cycle is related to higher tendency to suffer from severe injuries among elite female futsal players

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ABSTRACT

Objective: To describe the injury incidence according to each phase of menstrual cycle (MC) in professional futsal players along two consecutive season and, secondly, to compare the injury incidence and frequency between different phases of the MC.
Design: Prospective cohort study.
Settings: Elite futsal players.
Participants: 179 players.
Main outcome measure: Players were registered along two consecutive seasons.
Results: 191 injuries were reported throughout both seasons. The injury incidence was 30.63 days-off during the follicular phase, 23.6 during ovulatory phase and 17.59 days-off in luteal phase, showing higher incidence during the follicular phase. No statistical differences ($p > 0.05$) were reported for any variable comparing among the three phases of MC.
Conclusions: This study suggests the relevance to track the MC, but reduces its possible relationship or influence on the injury distribution during each phase of the MC. The knowledge of injury incidence, burden and etiology is a key factor to design injury prevention programs with the focus on the most common injuries, where MC could be included as a complementary factor.
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1. Introduction

The menstrual cycle (MC) is one of most underestimated variables which may influence sport performance on female athletes (Bruinvels, Burden, Brown, Richards, & Pedlar, 2016; de Jonge, 2003; Frankovich & Lebrun, 2000; Reilly, 2000). In this sense, a recent study with British elite and non-elite athletes showed that heavy menstrual bleeding (HMB) is common in elite athletes, and half of them perceived that MC affected their training and performance (Bruinvels et al., 2016). The differences in hormonal distribution during each phase could cause some alterations in performance factors, as muscular stiffness (Bell et al., 2009), balance (Dos Santos Andrade et al., 2017) or ligamentous laxity (Balachandar, Marciniak, Wall, & Balachandar, 2017). This is, the MC is one of the most important performance factors in women, but has been also signaled as an injury risk factor. Taking into account this, knowing more in deep the relationship of the MC with the injury incidence and burden could be a key factor to adapt prevention strategies and, linked to this, improve sport performance in female athletes.

Some studies reported differences in injuries regarding the phase of MC (Balachandar et al., 2017; Herzberg et al., 2017; Hewett, Zazulak, & Myer, 2007; Shultz, Wideman, Montgomery, & Levine, 2011). The main findings suggested a higher risk during the follicular phase, due to the increment of joint laxity, changes in neuromuscular control, static balance, and influence of motivational factors (Balachandar et al., 2017; Herzberg et al., 2017). Other

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authors showed a greater risk to suffer an injury during premenstrual and menstrual phases (Moller-Nielsen & Hammar, 1989). Furthermore, other studies reported controversial results in relation to the MC on sports performance and injury risk, showing higher performance in follicular or luteal phase, as well as higher injury risk in both main phases of the MC (Balachandar et al., 2017; García-Pinillos, Lago-Fuentes, Bujalance-Moreno, & Pérez-Castilla, 2020; Hewett et al., 2007; Romero-Moraleda et al., 2019). These findings suggest that more studies with better methodology are needed.

At present, futsal is a sport with increasing popularity, reaching a global television audience of almost 140 million people on the past 2012 FIFA Futsal World Cup in Thailand (Moore et al., 2018). Additionally, female futsal is also becoming more popular, celebrating the first UEFA Women's Futsal Eurocup in 2019. Particularly, in Spain, licenses have increased more than 35% in last 10 years (RFEF, 2018). However, literature about sport performance and injury epidemiology in this sport is still scarce (Lago-Fuentes et al., 2020). This is, the knowledge of the injury incidence has a clear relationship with the improvement on the sport performance due to the availability of players for the staff. Regarding the first point, futsal is signaled as a sport with injury rate among 5–10 injuries/1000 h of exposure time, being the ankle sprain the most common injury, both in training sessions and matches (Angoorani, Haratian, Mazaherinezhad, & Younespour, 2014; Gayardo, Matana, & Silva, 2012; Ruiz-Pérez et al., 2019; Serrano, Shahidian, Voser, & Leite, 2013). However, most of previous studies used a low quality research methodology (retrospective methodology; including both male and female players without a specific analysis for gender; without reporting exposure times (time of training and/or competing), or data were registered in amateur leagues. As it was previously explained, low injury rates (incidence and burden) are a key point to improve the sport performance. Regarding the female performance, gender-specific aspects of physiology (Costello, Bieuzen, & Bleakley, 2014; Herzberg et al., 2017; Hewett et al., 2007; Julian, Hecksteden, Fullagar, & Meyer, 2017) are key points to define and to train according to their importance. The MC is one of the most important factors in women, which has been also related an injury risk factor. However, none of previous studies took into account the distribution of injuries regarding different phases of MC in futsal teams.

To sum up, despite the increasing amount of scientific evidence about female performance in the last years, the MC and its effect on the athletic performance and injury risk remain still unknown. Only, a recent study performed a relevant research analyzing the injury incidence along the phases of MC in football players, but in lower categories (Martin et al., 2021). In this sense, to the best of the authors' knowledge, no previous studies have analyzed the relationship between menstrual phases and injury epidemiology (injury incidence, injury burden, injury etiology) in elite futsal players along two seasons. For these reasons, the aims of this study were to describe the injury incidence according to each phase of menstrual cycle in professional futsal players along two consecutive season and, secondly, to compare the injury incidence and frequency between different phases of the MC.

2. Methods

2.1. Study design

A longitudinal prospective cohort study was conducted during two consecutive seasons (2017–2018 and 2018–2019) with fourteen professional female futsal teams. 34 teams were contacted by the main researcher of the study with a presentation letter with a detailed dossier explaining the aims of the study, as well as with a detailed dossier with the variables and procedure. After this first contact, finally fourteen teams agreed to participate on this research. After that, staff exposed the main goals and the data which will be registered to the players, who signed the informed consent document according to the ethical standard in sport and exercise science (Harriss & Atkinson, 2015). The design and procedures of this study were approved by the Institutional Review Board (CEI 05/2020). During both seasons (since the pre-season to the last week of the season), staff was asked to register the different variables related to each injury event and, also, the workloads per microcycle. Regarding the MC, all players were asked to register their different phases by a calendar-based counting (as Google Calendar) or different apps. This registration was verified by the physical coaches during both seasons.

2.2. Participants

Fourteen teams who competed during 2017–2018 and 2018–2019 seasons in the Spanish First (11) and Second National Futsal (3) leagues agreed to participate voluntarily in this study. Four teams participated in both seasons; meanwhile three teams participated only in first season and other three in the second one. 195 players participated in these teams, but 16 players were excluded because they do not participate the whole season with their respective teams or because used hormonal contraceptive (HC). Finally, a total of 179 players-seasons (59.9 ± 7.8 kg; 164.2 ± 5.6 cm; Body Mass Index: 22.2 ± 2.3; 10.7 ± 5.1 years of experience) (see Table 1), conformed the final sample. This sample is bigger than the 90% for power of the sample size recommended for epidemiological studies (Hewett et al., 2007). These players trained 5–8 h and competed an average of one official match per week (30–35 matches per season). Regarding player position, they were categorized by 30 goalkeepers, 42 defenders, 78 wingers and 29 pivots. Finally, 155 were right-dominant and 24 left-dominant leg.

2.3. Procedure

2.3.1. Injury definition and variables

Injury was defined as: "an injury that occurs during a training session or match and which causes the player to be out of the next training session or match" (Hagglund, Walden, Bahr, & Ekstrand, 2005). For instance, accidents or injuries out of trainings and matches were excluded, as well as absences to training or matches due to illness or other reasons were not registered. Regarding this context, players were fully recovered from an injury when the medical staff confirmed they were prepared to train or compete without restrictions (Hagglund et al., 2005). The injury severity was categorized regarding the number of days-off: slight (1–3 days), minor (4–7 days), moderate (8–28 days) and major (>28 days) (Hagglund et al., 2005). Furthermore, different injury variables were registered, such as: type of injury, injury location, etiology, recurrence (Ruiz-Pérez et al., 2019), moment of the session, type of session, day of injury according to match and the quarter of the season (more information about operational definitions are displayed in **Appendix**).

2.3.2. Injury rates

Player exposure time is the time that players are involved in trainings and matches. Regarding this, the time involved in training sessions and matches (friendly and competitive games) was recorded daily in minutes by physical coaches, following the athletes at risk method (Lago-Fuentes et al., 2020; Stovitz & Shrier, 2012), including the pre-season and in-season periods. The athletes at risk methods consists on the number of players on the field

Table 1
Team, season characteristics and exposure time.

	17/18 Season	18/19 Season	Total	Mean (±SD)
Players	89	90	179	89.5 ± 0.5
Years of practice	11.5 ± 4.7	9.8 ± 5.4	–	10.7 ± 5.1
Height	164.4 ± 5.7	164.0 ± 5.6	–	164.2 ± 5.6
Body mass	59.9 ± 7.4	59.9 ± 8.2	–	59.9 ± 7.8
BMI	22.1 ± 2.1	22.2 ± 2.6	–	22.2 ± 2.3
Weeks	40	41	81	40.5 ± 0.5
Total Training time exposure (h)	20357.7	16573.5	36931.2	18465.6 ± 2675.8
Training time exposure (mean per team) (h)	2908.2	2367.6	–	2637.9 ± 657.9
Total Match time exposure (h)	873.3	913.3	1786.7	893.3 ± 28.3
Match time exposure (mean per team) (h)	124.8	130.5	–	127.6 ± 13.0
Total time exposure (h)	21231.0	17486.8	38717.8	1938.9 ± 2647.6
Total time exposure (mean per team) (h)	3033.0	2498.1	–	2765.6 ± 661.4

multiplied by either the number of games or the number of game-hours that the team has played (Stovitz & Shrier, 2012). According to this, the training exposure time is considered the sum of sessions by their mean duration, multiplied by number of players per team, with also preseason and in-season periods (Lago-Fuentes et al., 2020). Second, the match exposure time was calculated as the duration of the official matches (40 min of stopped time/60) by number of athletes playing at the same time on the court (five players), including also friendly matches (Lago-Fuentes et al., 2020). With these data, injury prevalence, injury incidence and injury burden were calculated. The injury prevalence was rated as the number of players injured split by number of available players (Knowles, Marshall, & Guskiewicz, 2006). The overall injury incidence, match injury incidence and training injury incidence were the number of injuries divided by 1000 player-hours in total, match and training, respectively. Then, the injury burden was calculated as the number of days off per each 1000 h (Bahr, Clarsen, & Ekstrand, 2018).

2.3.3. Menstrual cycle measurement

At the beginning of both seasons, each physical coach explained the relevant information about the MC, the methods to self-register the length of the MC and how to split it in the main phases of the MC. In this meeting, staff explained the main categories to register the MC, which are the following: follicular (1–12 days), ovulatory (13–15 days), luteal (16–28 days), considering the players having a regular MC of 28 days (de Jonge, 2003; Hamilton, 2012; Reilly, 2000), always starting the cycle with the first day of bleeding. Taking into account the variability on the length of MC in athletes, each player was encouraged to calculate their three main phases regarding their mean length of MC after a meeting to explain how to calculate them, associated to the main milestones of the MC. Regarding this, players self-registered their own MC using calendar-based counting as well as different apps. Finally, the use of HC was registered during the second season as exclusion criteria. Regarding this, only a 5% of the players were excluded because of its use.

2.4. Statistical analysis

All data collected along two consecutive seasons were unified in a database (Excel for Windows). Then, the database was categorized and codified with SPSS for Windows (20.0, Chicago, IL, USA). Descriptive data are presented as means and standard deviation, proportions (%), incidence rates and 95% confidence intervals (CI) (Knowles et al., 2006). For comparing the differences on injuries between MC phases, one-way ANOVA and chi-squared test were conducted according the nature of variables, when appropriate. Statistical level was set at $p < 0.05$.

3. Results

3.1. Exposure time, incidence rate and injury burden

A total of 191 injuries occurred during both seasons, 104 during training sessions and 87 during matches (see Table 2). The injury incidence and injury burden were higher during official and recreational competitions compared with training sessions. Regarding the comparison among different phases of MC, no statistical differences were observed on injury incidence neither injury burden between them (Table 2) ($p > 0.05$ in all comparisons). However, clinical differences were observed among the three phases, with follicular phase showing the highest time lost by an injury (30.63 days-off) compared with ovulatory and luteal phases, with 23.60 and 17.59 days-off, respectively.

3.2. Type, location, type of session and menstrual cycle

The most common injuries regarding location were in quadriceps (22.0%) and ankle (20.4%) (see Table 3). Regarding the comparison among phases, there were no statistical differences on the distribution of injuries among follicular, ovulatory and luteal phases ($p > 0.179$) for all anatomy locations.

The most common type of injuries were strain (40.8%) and sprain (27.7%), followed by contusions (9.9%). However, no statistical differences were observed comparing their distribution among the three different phases of MC ($p > 0.395$).

3.3. Place, temporal distribution and menstrual cycle

Table 4 showed the temporal distribution of injuries regarding the moment of the session (second half: 44.0%), the match-day approach (MD+2: 10.5%) and the moment of the season by quarters (Q1: 44.0%). Regarding phases of the MC, no differences between phases was reported for three factors (p: 0.424, 0.161 and 0.380, respectively). Only a tendency of frequencies showed more injuries in follicular phase during the second half of the session, MD-2, MD+2 and for all the quarters.

4. Discussion

This prospective cohort study showed an injury incidence of 5.68 injuries/1000 h of exposure time and an injury burden of 9.32 days-off. Regarding the distribution of injuries, we could observe that there were not statistical differences in any ratio neither variable comparing the phases of the MC: follicular, ovulatory and luteal. Only some tendencies of higher frequencies of injuries were reported in follicular phase compared with other two stages.

Table 2
Injury incidence and injury burden by menstrual phase.

	Total				Follicular				Ovulatory				Luteal				P-values	
	Hours exposure	Injuries	Incidence (95% CI)	Injury burden	Injuries	Incidence (95% CI)	Injury burden	Injuries	Incidence (95% CI)	Injury burden	Injuries	Incidence (95% CI)	Injury burden	Injury incidence	Injury burden
Training	**36931.13**	**104**	**3.52 (3.27–3.77)**	**9.79**	**58**	**3.25 (2.98–3.52)**	**11.44**	**16**	**3.20 (2.75–3.63)**	**9.28**	**30**	**3.45 (3.11–3.80)**	**7.65**	p > 0.05	p > 0.05
Slight	24	3.81 (3.23–4.39)	0.82	9	3.98 (2.85–5.11)	0.85	3	4.06 (2.83–5.30)	1.10	12	3.63 (2.95–4.31)	0.72			
Minor	33	3.64 (3.16–4.12)	2.24	19	3.42 (2.78–4.08)	2.37	3	4.12 (2.94–5.31)	2.09	11	3.88 (3.11–4.66)	2.06			
Moderate	35	3.08 (2.68–3.48)	5.24	22	2.98 (2.42–3.55)	5.18	6	3.10 (2.61–3.59)	6.23	7	3.36 (2.71–4.00)	4.56			
Major	12	3.91 (3.16–4.67)	42.89	8	4.04 (3.25–4.84)	43.87	4	3.67 (2.29–5.06)	40.94	–	–	–			
Match	**1786.67**	**87**	**58.38 (53.98–62.78)**	**196.82**	**37**	**54.13 (49.74–58.53)**	**240.07**	**14**	**58.86 (49.26–68.46)**	**185.98**	**36**	**51.04 (45.23–56.84)**	**139.51**	p > 0.05	p > 0.05
Slight	8	60.51 (46.93–74.08)	15.68	2	64.59 (17.45–111.72)	14.89	1	48.84 (–)	21.87	5	61.21 (51.23–71.18)	15.73			
Minor	24	56.64 (47.93–65.36)	40.61	9	54.87 (44.92–64.83)	43.91	4	64.24 (44.23–84.25)	37.89	11	55.32 (40.36–70.28)	40.71			
Moderate	35	62.92 (55.67–70.19)	110.89	17	66.01 (56.10–75.92)	112.92	5	73.58 (65.71–81.46)	130.72	13	54.78 (42.35–67.20)	111.82			
Major	20	51.68 (41.97–61.38)	843.91	9	50.09 (38.71–61.49)	897.68	4	68.32 (50.59–86.04)	730.58	7	44.20 (27.01–61.39)	840.78			
Total	**38717.79**	**191**	**5.68 (5.41–5.96)**	**9.32**	**95**	**5.64 (5.24–6.04)**	**10.89**	**30**	**5.59 (4.95–6.23)**	**8.79**	**66**	**5.79 (5.29–6.29)**	**7.24**	p > 0.05	p > 0.05
Slight	32	5.44 (4.73–6.12)	0.82	11	5.67 (4.47–6.88)	0.77	4	5.80 (4.53–7.08)	1.05	17	6.09 (5.16–7.02)	0.80			
Minor	57	5.89 (5.22–6.57)	2.07	28	5.83 (4.98–6.69)	2.18	7	5.30 (3.67–6.93)	1.66	22	5.77 (4.87–6.66)	2.06			
Moderate	70	5.41 (4.80–6.01)	4.82	39	5.51 (4.96–6.05)	4.71	11	5.49 (4.95–6.57)	5.07	20	5.76 (4.95–6.57)	4.92			
Major	32	6.75 (5.47–8.02)	40.45	17	5.58 (4.68–6.50)	45.99	8	5.88 (4.38–7.37)	24.02	7	5.23 (3.31–7.15)	45.79			

CI: Confidence interval.

4.1. Exposure time, incidence rate and injury burden

The follicular phase has been signaled as the most sensitive period to suffer an injury in previous studies (Balachandar et al., 2017; Chidi-Ogbolu & Baar, 2019; Herzberg et al., 2017; Hewett et al., 2007; Tourville et al., 2016). Regarding this, previous studies suggested that modifications in ligamentous laxity during follicular phase can be as one of the main injury risk factors (de Jonge, 2003; Reilly, 2000). This is more relevant when anterior cruciate ligament (ACL) injuries are analyzed (Dos Santos Andrade et al., 2017), due to the loss of hamstring-quadriceps peak torque strength balance during this phase (Beynnon et al., 2006; Hewett et al., 2007; Martin et al., 2021; Ruedl et al., 2009). Notwithstanding, there is no consensus about the distribution of these injuries along the MC, as a recent meta-analysis reported, with a major tendency of suffering ACL injury during the ovulatory phase (Herzberg et al., 2017). Our findings are in line with this controversy, with no statistical differences on injury incidence neither injury burden comparing follicular, ovulatory and luteal phases. However, a higher tendency of incidence rates was observed in follicular phase compared with other ones. One of the hypothetical reason of these contradicting results may be based on the controversy regarding the designation of each menstrual phase terminology (i.e, different names for the same phase: bleeding or menstrual) and the duration of each phase linked with different names (Zazulak, Paterno, Myer, Romani, & Hewett, 2006). An example of this controversy may be reflected in the classification of the follicular phase in the present study and previous researches (Lee et al., 2013). For instance, we classified the follicular phase 1–12 day of MC but in several studies this period of time was split in two (Beynnon et al., 2006; Lee et al., 2013) or even the duration of the follicular phase was established in 1–9 days (Heitz, Eisenman, Beck, & Walker, 1999). Consequently, it is difficult to compare the results between the different studies whether they are based on the days of the MC. However, our study follows the last recommendations about the division of the MC, which highlights the relevance of our findings (CITA MCNULTY). For these reasons, further studies following this classification and measuring physiological variables (such as estrogen and progesterone concentrations with ovulatory test or blood extraction) are needed to compare the injury incidence and injury burden among these three phases of MC.

4.2. Type, location, type of session and menstrual cycle

Quadriceps and ankle were the most affected locations, meanwhile knee injuries were the third most common (16.8%) but with more days-off, which happened mainly during follicular phase. In this regard, Wojtys and colleagues (Wojtys, Huston, Boynton, Spindler, & Lindenfeld, 2002) showed in their study an increase higher than 2.5 times in the ACL ruptures in the period of time where the estrogen rise (9–14 day of MC) vs follicular (1–8 day of MC) or luteal (15–28 day of MC) phases. This tendency of suffering from tendon and ligament injuries during the rise of estrogen may be partially explained for the influence of these sex hormones in the loss of mechanical properties. The increase of estrogen, estradiol in particular, has a direct impact on the tendon and ligament, becoming these structures stiffer (Shultz et al., 2011). In our study, a relative increase in the frequency and duration of these injuries is shown during this follicular phase, but without statistical significance, as it has been mentioned before, the rise in estrogen levels could explain partially a higher likelihood of suffer from sinew injury in this phase. However, more studies will analyze hormone levels to ensure this possible relationship among injuries during one phase and its link to estrogen levels.

Table 3
Injury location, type of injuries and type of the session according to the phase of menstrual cycle.

Injury location	Total (%)	Follicular (%)	Ovulatory (%)	Luteal (%)	p-value
Quadriceps	42 (22.0)	21 (11.0)	7 (3.7)	14 (7.3)	0.179
Ankle	39 (20.4)	26 (13.6)	2 (1.0)	11 (5.8)	
Knee	32 (16.8)	14 (7.3)	9 (4.7)	9 (4.7)	
Hip/Groin	24 (12.6)	11 (5.8)	4 (2.1)	9 (4.7)	
Shoulder/Arm	10 (5.2)	6 (3.1)	–	4 (2.1)	
Hamstrings	7 (3.7)	(1.6)	–	4 (2.1)	
Back	8 (42)	3 (3.2)	–	5 (2.6)	
Calf	9 (4.7)	4 (2.1)	3 (1.6)	2 (1.0)	
Foot	8 (4.2)	2 (1.0)	1 (0.5)	5 (2.6)	
Head	2 (1.0)	1 (0.5)	1 (0.5)	–	
Other	10 (5.2)	4 (2.1)	3 (1.6)	3 (1.6)	
Type of injury	**Total (%)**	**Follicular (%)**	**Ovulatory (%)**	**Luteal (%)**	**p-value**
Strain (Muscle)	78 (40.8)	34 (17.8)	14 (7.3)	30 (15.7)	0.395
Sprain (Lig)	53 (27.7)	32 (16.8)	7 (3.7)	14 (7.3)	
Rupture (Lig)	9 (4.7)	6 (3.1)	–	3 (1.6)	
Rupture (Muscle)	10 (5.2)	7 (3.7)	1 (0.5)	2 (1.0)	
Contusion	19 (9.9)	6 (3.1)	5 (2.6)	8 (4.2)	
Other	22 (11.5)	10 (5.2)	3 (1.6)	9 (4.7)	
Type of session	**Total (%)**	**Follicular (%)**	**Ovulatory (%)**	**Luteal (%)**	**p-value**
Conditioning	26 (13.6)	12 (6.3)	2 (1.0)	12 (6.3)	0.066
Tactical	78 (40.8)	46 (24.1)	14 (7.3)	18 (9.4)	
Match	87 (45.5)	37 (19.4)	14 (7.3)	36 (18.8)	

Table 4
Temporal distribution of injuries according to the phase of menstrual cycle.

Moment	Total (%)	Follicular (%)	Ovulatory (%)	Luteal (%)	p-value
Warm-Up	19 (9.9)	9 (4.7)	3 (1.6)	7 (3.7)	0.424
First Half	50 (26.2)	29 (15.2)	5 (2.6)	16 (8.4)	
Second Half	84 (44.0)	40 (20.9)	18 (9.4)	26 (13.6)	
End Session	38 (19.9)	17 (8.9)	4 (2.1)	17 (8.9)	
MD[a]	**Total (%)**	**Follicular (%)**	**Ovulatory (%)**	**Luteal (%)**	**p-value**
MD-4	7 (3.7)	4 (2.1)	3 (1.6)	–	0.161
MD-3	11 (5.8)	7 (3.7)	2 (1.0)	2 (1.0)	
MD-2	15 (7.9)	8 (4.2)	3 (1.6)	4 (2.1)	
MD -1	5 (2.6)	4 (2.1)	–	1 (0.5)	
MD	87 (45.5)	38 (19.9)	15 (7.9)	34 (17.8)	
MD+1	14 (7.3)	11 (5.8)	–	3 (1.6)	
MD+2	20 (10.5)	12 (6.3)	2 (1.0)	6 (3.1)	
MD+3	7 (3.7)	3 (1.6)	1 (0.5)	3 (1.6)	
MD+4	5 (2.6)	1 (0.5)	2 (1.0)	2 (1.0)	
No MD	20 (10.5)	7 (3.7)	2 (1.0)	11 (5.8)	
Quarter[a]	**Total (%)**	**Follicular (%)**	**Ovulatory (%)**	**Luteal (%)**	**p-value**
Q1	84 (44.0)	43 (22.5)	9 (4.7)	32 (16.8)	0.380
Q2	40 (20.9)	16 (8.4)	10 (5.2)	14 (7.3)	
Q3	38 (19.9)	20 (10.5)	5 (2.6)	13 (6.8)	
Q4	29 (15.2)	16 (8.4)	6 (3.1)	7 (3.7)	

[a] MD: match day; Q1: First quarter; Q2: second quarter; Q3: third quarter; Q4: fourth quarter.

We discovered that two of each three sprains happened during the follicular phase. Regarding this finding, no previous studies compared this type of injuries along the three phases. This could be related to the increase of stiff in sinew is the rise in joint laxity which implies more instability in the first half of the cycle. That is, a previous study reported a positive correlation between knee laxity and estrogen levels (Park, Stefanyshyn, Loitz-Ramage, Hart, & Ronsky, 2009; Shultz et al., 2011). The partial inhibition of lysyl oxidase activity, due to a high estrogen milieu, could explain the decreasing of the fiber enzymatic cross-linking and hence the loss of mechanical force production. For this reason, this distribution should be taken into account to plan different preventive strategies to improve motor control and stiffness to counteract the laxity due to low values of estrogen and progesterone, especially during the follicular phase (Chidi-Ogbolu & Baar, 2019; Hewett et al., 2007; Slauterbeck et al., 2002; Tourville et al., 2016). Also, due to the small number of studies focused in the comparison of injuries in each MC phase, further studies should focus on this type of injuries to other locations, at least, for elite female futsal players.

4.3. Place, temporal distribution and menstrual cycle

Regarding temporal distribution of injuries along the MC, previous studies did not report results about the relation between type of session, day of the week, or the quarter of the season. The moment of the season could be considered as a risk factor, with more injury incidence during preseason in comparison with the rest of the season (Lago-Fuentes et al., 2020). We also discovered more injuries during the first quarter of the season, which can be related to the high volume of workouts during the first weeks of the season. Also, the moment of the session (beginning or the end) could be considered as a risk factor, especially due to the fact of the fatigue accumulated along the training session or the minutes played during the match. We found that more than 60% of injuries happened from the second half to the end of the sessions, which suggests the relevance of the fatigue on the sessions (McCall et al., 2014; Ruiz-Pérez et al., 2020). However, regarding the aims of this study, we showed more incidence during the follicular phase in the first quarter and at the end of the sessions (that is, in more fatigued situations), but without statistical differences in these variables along the phases of MC, maybe due to the number of injuries registered or the methods to register these phases.

To the authors' knowledge, this is the first prospective study which has registered the distribution of injuries during two seasons along the phases of MC in elite team sport players, in this case, futsal teams. However, some limitations need to be addressed. First of all, the MC phases were recorded without the control of hormone concentrations, which should be the most objective method (Janse DE Jonge, Thompson, & Han, 2019). Nevertheless, recent studies have accepted this limitation in cohort studies, due to the fact that the absence of hormonal control favors a larger sample (Martin et al., 2021), as well as the expensiveness of controlling hormone levels along a whole season in a cohort sample. Second, during the

first season, the use of HC was not controlled. In this sense, the second season reported about a 5% of players using HC (which were discarded for this study), which can indicate the low frequency of HC consumption in this sport. However, further studies should include players with HC intake in epidemiological studies to know its influence on injury distribution. . Finally, the exposure time was calculated indirectly, but this bias has been also accepted due to the trouble and logistic to register this individual data per player in different teams (Stovitz & Shrier, 2012). Also, there was difference in the total exposure time among both seasons, but this was absorbed when injury incidence and injury burden were calculated. Linked to these limitations, further studies should include more teams of each category to analyze if there are differences on injuries according to the competitive level, as well as registering the hormonal concentrations to ensure the objective categorization of injuries regarding their phases of MC (Janse DE Jonge et al., 2019) and including athletes with HC consumption.

5. Conclusions

The main finding of the present study suggests a larger tendency of elite female futsal players to suffer an injury during the follicular phase, with more severity and, consequently, more days-off. Interestingly, ligament injuries reported higher incidence also during the follicular phase. This situation might be partially explained by the estrogen milieu and their implication in the loss of mechanical properties of the tendon. However, more studies are needed to conclude this relationship. Finally, the distribution of injuries along the session and the season did not report more frequency in a specific phase of MC, but the severity of them could be linked to different phases. The information provided in this manuscript could be of interest for sport scientist and physical coaches to track the menstrual cycle of their athletes and improve the wellbeing of them along the season with the focus on improving the neuromuscular adaptations to reduce muscular and ligament injuries along the season, mainly at the end of the sessions and at the beginning of the season.

To sum up, these findings are relevant for physical coaches to understand the relevance to monitor the menstrual cycle on their athletes and, also, educate them to know in deep their side effects on each phase to, consequently, try to modulate training loads (volume, intensity, frequency …), as well as modifying physical characteristics to indicate to coaches, among other strategies. Furthermore, this procedure is easy to apply and help both figures (players and technical staff) to understand one of the most underestimated influencing factors on sport performance and injury risk. Finally, our findings could serve to develop further studies comparing the injury incidence and etiology after different types of preventive programs, modulating the loads regarding each MC phase.

Ethical statements

The institutional ethics committee of the first author' university approved this study (CEI 05/2020).

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Declaration of competing interest

None declared.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ptsp.2021.08.008.

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